Eidgenössische Technische Hochschule Zürich

# Physik III

Autumn Semester 2026

Lecture Notes

# Contents

**1 Optics** **1**
1.1 Fermats Principle . . . 1
1.2 Lenses . . . 2
1.3 Wave Optics . . . 3
1.4 Polarisation of Light . . . 5
1.5 Spectroscopy . . . 7

**2 Photons** **11**
2.1 Photoelectric effect . . . 11

**3 Atoms and their Composition** **13**

**4 Statistical Mechanics** **14**
4.1 The Boltzmann Distribution . . . 14

# References

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# Physik III

Fynn Krebser–fkrebser@student.ethz.ch

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Lec 1

## 1 Optics

Optics is the study of light. It is relevant as light is everywhere. Furthermore it is a key tool for measurements in science. Furthermore, it introduces ideas associated with quantum mechanics. Historically, there was a wave and a corpuscle[1] picture of light. This wave/particle duality is the essence of quantum mechanics.

### 1.1 Fermats Principle

Fermat's principle tries to describe the path of light.

**Proposition 1.1: Fermat's Principle**

There are two different statements of Fermat's principle.

"Light travels by the route which takes the shortest time."

"Light will take the path which is extremal with respect to variations in the optical path"

The first statement is more intuitive, but the second statement is more general. For example, consider a setup with a mirror

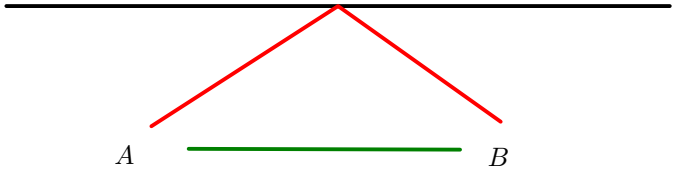



Figure 1: Two people seeing each other with a mirror

**Definition 1.2: Optical Path**

The OPTICAL PATH is a measure of time taken, written as distance.

$$D = ct = \frac{cs}{v}.$$

Where $c$ is the speed of light and $t$ is the time taken.

Recall that the speed of an electromagnetic wave in a medium is given by $v = \frac{c}{n}$, where $n$ is the refractive index of the medium. Thus we can write the optical path as

$$D = \frac{cs}{v} = ns.$$

[1] Light comes as particles

If the light has a certain frequency $f$, it will oscillate ever $t_c = \frac{1}{f}$. So the number of cycles in a time $t$ is given by

$$N = \frac{t}{t_c} = \frac{fD}{c} = \frac{D}{\lambda_0},$$

where $\lambda_0$ is the FREE-SPACE WAVELENGTH of light.

The optical path tells us how many cycles of oscillation we have convered on our path.

We now want to discover the refraction law using fermat's principle.

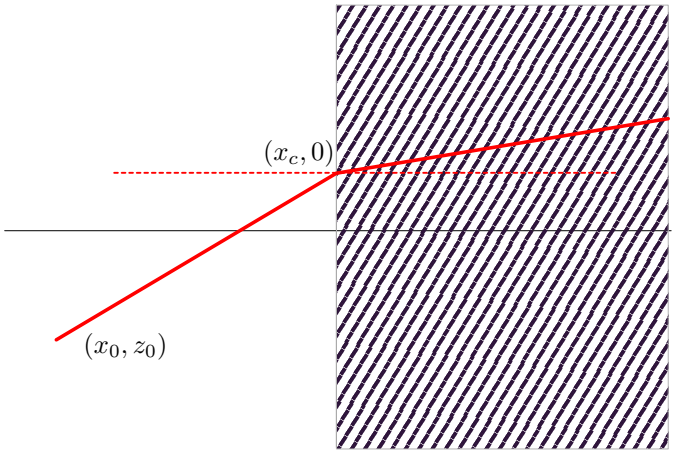



Figure 2: Deriviation of Snell's law using Fermat's principle

The optical path in the left half is given by

$$D_1 = n_1\sqrt{(x_c - x_0)^2 + z_0^2}).$$

In the right half, we have

$$D_2 = n_2\sqrt{(x_p - x_c)^2 + z_p^2}).$$

So we want to minimize the total optical path $D = D_1 + D_2$ with respect to $x_c$. We have

$$\frac{dD}{dx_c} = \frac{dD_1}{dx_c} + \frac{dD_2}{dx_c} = 0$$

$$\frac{n_1}{\sqrt{(x_c - x_0)^2 + z_0^2}}(x_c - x_0) = \frac{n_2}{\sqrt{(x_p - x_c)^2 + z_p^2}}(x_p - x_c)$$

$$n_1 \sin(\theta_1) = n_2 \sin(\theta_2).$$

Where $\theta_1$ and $\theta_2$ are the angles of incidence and refraction, respectively. This is known as Snell's law.

Theorem 1.3: **Snell's Law**

A light ray incident on a boundary between two media with refractive indices $n_1$ and $n_2$ will be refracted according to the law

$$n_1 \sin(\theta_1) = n_2 \sin(\theta_2).$$

## 1.2 Lenses

The best way to think of a lens is to think about the eye. It consists in principle of a lens and a screen (the retina). In principle, the lens is of refractive index $n_2$ in some sort of substance of refractive index $n_1$. We are restricted by the shape and the refractive indices of the lens. Furthermore, we are restricted by the size of the lens.

Let us focus on the ideal lens.

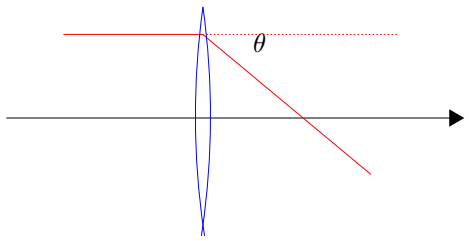



Figure 3: Ray bending of an ideal lens.

Light hitting the lens at a height $h$ is deflected by the angle $\theta = \frac{h}{f}$, where $f$ is the FOCAL LENGTH of the lens. Light incoming parallel to the optical axiw will cross the optical axis at a distance $f$ after the Lens. The intersection is called FOCAL POINT. Here we make use of the PARAXIAL APPROXIMATION, which states that we only consider rays close to the optical axis, so we can use

$$\sin(\theta) \approx \tan(\theta) \approx \theta.$$

We will see, that in the paraxial approximation, if we have a non parallel ray incoming, it will still focus a distance $f$ after the lens. In general, parallel rays converge after the lens in the FOCAL PLANE.

We now want to trace rays through a single lens.

By similar triangles, we have

$$\frac{\Delta}{\delta} = \frac{d}{s} =: M.$$

where $M$ is the MAGNIFICATION of the lens. By similar triangles again, we have

$$\frac{\delta}{f} = \frac{\Delta}{d-f} \Rightarrow \frac{d-f}{f} = \frac{\Delta}{\delta} = \frac{d}{s}.$$

Hence,

$$\frac{d}{f} - 1 = \frac{d}{s} \Rightarrow \frac{1}{f} = \frac{1}{s} + \frac{1}{d}.$$

This is called the LENS EQUATION.

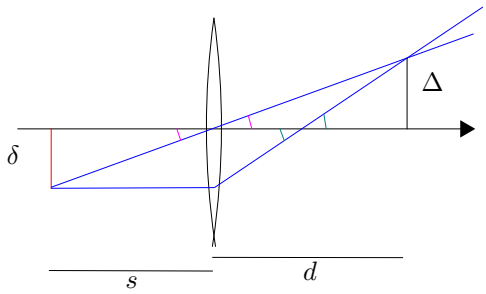



Figure 4: Deriviation of the lens equation using ray tracing.

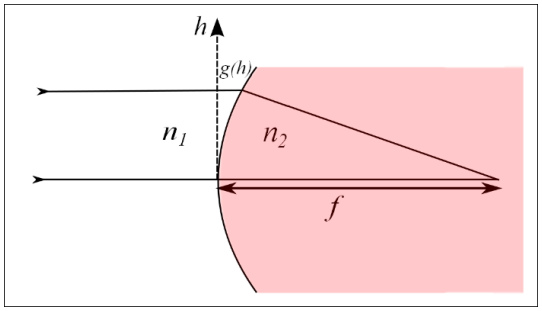



Figure 5: Setup of the surface lens.

Let us figure out how to design a simple "surface" lens. For this, we describe the profile by $g(h)$.

The optical path is given by

$$D = n_1 g(h) + n_2 \sqrt{h^2 + (f - g(h))^2}.$$

In general this is not really solvable, so we consider the paraxial approximation. So we have $h \ll f$. Since the lens should be symmetric we can assume

$$g(h) = \sum_n a_{2n} h^{2n}.$$

The simplest solution would be $g(h) = a_2 h^2$. Putting this into the expression, we get

$$\begin{aligned} D &= n_1 a_2 h^2 + n_2 \sqrt{h^2 + (f - a_2 h^2)^2} + O(h^4) \\ &= n_1 a_2 h^2 + n_2 \sqrt{h^2 + f^2 + a_2^2 h^4 - 2 f a_2 h^2} + O(h^4). \end{aligned}$$

Using that $f \gg a_2 h^2, h$ we get

$$D = n_1 a_2 h^2 + n_2 f \sqrt{\frac{h^2}{f^2} + 1 + \cancel{\frac{a_2^2 h^4}{f^2}}^{\text{neglect}} - \frac{2 a_2 h^2}{f}} + O(h^4).$$

Using the taylor expansion $\sqrt{1+x} = 1 + \frac{x}{2} + O(x^2)$, we get

$$D = n_1 a_2 h^2 + n_2 f \left(1 + \frac{h^2}{2f^2} - \frac{a_2 h^2}{f}\right) + O(h^4).$$

Reordering the terms, we get

$$D = n_2 f + \left(n_1 a_2 + \frac{n_2}{2f} - n_2 a_2\right) h^2 + O(h^4).$$

We want all paths to have the same optical path. So we want the coefficient of $h^2$ to vanish. This gives us

$$n_1 a_2 + \frac{n_2}{2f} - n_2 a_2 = 0 \Rightarrow a_2 = \frac{n_2}{2f(n_2 - n_1)}.$$

Unfortunately it is very annoying to make a lens with a parabolic profile. A spherical profile would be much easier.

Lenses in general are imperfect. Possible sources are:

1. Ignored higher order terms: symmetric about axis ⇒ spherical aberration

2. Off axis aberrations: coma.

3. Chromatic aberrations: The refractive index depends on the wavelength of light. So different colors focus at different points.

Lec 2

Let us look at some imaging instruments in more detail. The human eye is limited by: the size of the eye is fixed. As we have seen

$$\frac{1}{f} = \frac{1}{s} + \frac{1}{d}.$$

For the human eye, $d$ is fixed. Secondly, the range of focal lengths is limited. Let us consider the telescope, using lenses to help. The main problem we want to solve is that we want to see two stars subtending a small angle $\theta$.

If we have only one lens, since the angle is small, we have

$$\theta = \frac{\delta}{s} = \frac{\Delta}{d}.$$

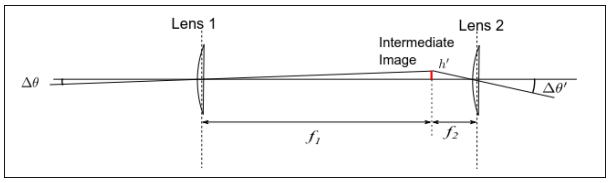



Figure 6: Telescope with two lenses.

Introducing a second lens, we first see, that image 1 is at $f_1$ behind lens 1, since

$$\frac{1}{f_1} = \frac{1}{s} + \frac{1}{d} \approx \frac{1}{d},$$

since $s$ is very large. For the second lens, we then find

$$\theta_2 = \frac{\Delta}{f_2} = \frac{d\theta}{f_2} = \frac{f_1}{f_2}\theta.$$

Nowadays, we use a telescope with a mirror instead of a lens. The advantage is that we don't get chromatic aberration, since the mirror reflects all wavelengths the same way.

For looking at very small things, we typically use a microscope. Recall the magnification

$$M = \frac{\Delta}{\delta} = \frac{d}{s}.$$

So if we want to see very small things, we need to make $s$ very small. The problem is that we need to satisfy the lens equation.

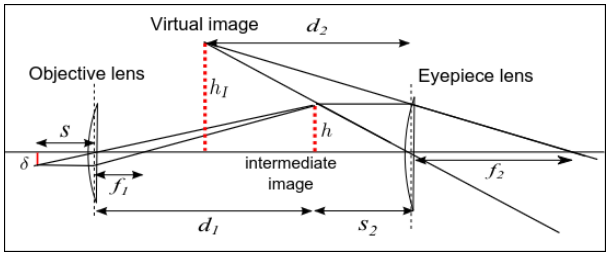



Figure 7: Microscope with two lenses.

Using a lens with $f_1 < s$ we thus get a magnification

$$h = \frac{d}{f_1}\delta.$$

In general, propagating through lenses is done, by finding the image through the first lens and then propagate that image through the second lens.

Typically in a microscope, we will use a lot of lenses. This is mainly because then we can only change the angle a little bit and hence reduce the spherical aberration. It also helps out cancelling out chromatic aberration.

## 1.3 Wave Optics

Starting from Maxwell's equations, we can turn them into a wave equation

$$-\nabla^2 \mathbf{E} = -\mu_0 \varepsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}.$$

To simplify the problem, we can decide to treat directions of $\mathbf{E}$ separately. Then for example,

$$-\frac{\partial^2 E_x}{\partial x^2} = -\mu_0 \varepsilon_0 \frac{\partial^2 E_x}{\partial t^2}.$$

This is the scalar wave equation.

Measuring the light typically involves measuring the intensity, given by

$$I = \frac{cn\varepsilon_0 |\mathbf{E}|^2}{2},$$

which is measured as power per unit area.

Let us look at the solutions of the scalar wave equation. It is given by

$$\nabla^2 V - \frac{1}{c^2}\frac{\partial^2 V}{\partial t^2} = 0.$$

We will use spherical coordinates, where the general solution is given by

$$V(r,t) = \frac{a}{r}\cos(kr - \omega t) + \frac{b}{r}\cos(kr + \omega t).$$

Here the first part represents an outgoing wave and the second part represents an incoming wave. Using complex numbers, we can write this as

$$V(r,t) = \frac{a}{r}e^{i(kr-\omega t)} + \frac{b}{r}e^{i(kr+\omega t)}.$$

We select the first term for a light source at the origin. Keep in mind that we can use complex numbers, but we can only measure real quantities. So we should always

finish with $\mathrm{Re}(V)$ or $|V|^2$. We now want to look at the propagation of light through an aperture.

For this we consider a point source at $(x_s, y_s, z_s)$. We then place a screen at $z = 0$ with a hole in it. We then want to find the field at a point of interest $(x_p, y_p, z_p)$.

Since we have a point source, the outgoing wave, emitted by the point source is given by

$$V_{\text{emit}} = A\frac{e^{i(ks-\omega t)}}{s},$$

where $s = \sqrt{(x-x_s)^2 + (y-y_s)^2 + (z-z_s)^2}$. To propagate this wave through the hole, we use HUYGEN'S PRINCIPLE.

**Proposition 1.4: Huygen's principle**

Given an incoming wave, we can write down the wavefronts. Letting each point on the wavefront be a point source of a new wave, we can then find the outgoing wave by summing up all the contributions of the new waves.

For example a planar wave front, for each point on the wave front, we can consider a point source. The outgoing wave is then given by the sum of all the contributions of the point sources. Summing up this results in again a planar wave front. As a recipe, first take the original wave

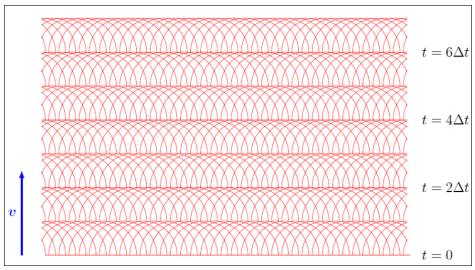



Figure 8: Huygen's principle for a planar wave front.

amplitude. The use this as a source of an outgoing wave at every position.

$$A\frac{e^{iks(x,y,z)}}{s(x,y,z)} \cdot \frac{e^{ikr}}{r},$$

where

$$r = \sqrt{(x_f - x)^2 + (y_f - y)^2 + (z_f - z)^2}.$$

Then, we sum up all the contributions of the outgoing waves.

For our problem, we constrain ourself to $x_f = x_p$, $y_f = y_p$ and $z_f = z_p$. Splitting the aperture into small pieces $\mathrm{d}x\mathrm{d}y$, we can write down the outgoing wave as

$$A\frac{e^{iks}}{s}\mathrm{d}x\mathrm{d}y \cdot \frac{e^{ikr}}{r}.$$

Summing up all the contributions, we get

$$U(P) \propto \iint \frac{e^{iks}}{s}\frac{e^{ikr}}{r}\mathrm{d}x\mathrm{d}y.$$

A more rigorous derivation would yield

$$U(P) = -\frac{iA}{4\pi}\oint \frac{e^{ik(s+r)}}{sr}\left(k\hat{r}\cdot\hat{n} - k\hat{s}\cdot\hat{n}\right)\mathrm{d}a.$$

Let us simplify the expression from Huygen's approach. First, since the aperture is small, compared to the distance to the source, we can approximate the amplitude as constant. I.e. if $s_0 \gg \lambda, \Delta_x, \Delta_y$, we can approximate $\frac{1}{s} \approx \frac{1}{s_0}$. So the incoming wave is given by

$$\frac{1}{s_0}e^{iks}.$$

Furthermore if the point of interest is far enough away, we can make the same approximation. Thus,

$$U(P) \propto \frac{A_0}{r_0 s_0}\iint e^{ik(s+r)}\mathrm{d}x\mathrm{d}y.$$

Defining the center of the aperture as $(0,0,0)$, we can write down the distances as

$$s_0 = \sqrt{x_s^2 + y_s^2 + z_s^2}, \quad r_0 = \sqrt{x_p^2 + y_p^2 + z_p^2}.$$

The interpretation of the integral is given as the amplitude is given by sum over amplitudes due to all paths. For this we simplify the expression. In general, the aperture size $\Delta_x, \Delta_y \ll |s_0|, |r_0|$. We have

$$\begin{aligned} r &= \sqrt{(x_p - x)^2 + (y_p - y)^2 + (z_p - z)^2} \\ &= \sqrt{x^2 + y^2 + z^2 + x_p^2 + y_p^2 + z_p^2 - 2(xx_p + yy_p + zz_p)} \\ &= \sqrt{x^2 + y^2 + r_0^2 - 2(xx_p + yy_p)} \\ &= r_0\sqrt{1 + \frac{x^2+y^2}{r_0^2} - 2\frac{xx_p + yy_p}{r_0^2}}. \end{aligned}$$

Using $x, y \ll r_0$, we get $\frac{x}{r_0} \ll 1 \Rightarrow \frac{x^2}{r_0^2} \lll 1$. Thus

$$r \approx r_0\left(1 - \frac{xx_p + yy_p}{r_0^2}\right).$$

Analogous calculation can be done for $s$. Thus we get

$$s \approx s_0\left(1 - \frac{xx_s + yy_s}{s_0^2}\right).$$

Hence,

$$U(P) = K\int_A e^{ik(s+r)}\mathrm{d}a = Ke^{ik(s_0+r_0)}\int_A e^{-ik(xX+yY)}\mathrm{d}a,$$

where

$$X = \frac{x_s}{s_0} + \frac{x_p}{r_0}, \quad Y = \frac{y_s}{s_0} + \frac{y_p}{r_0}.$$

The approximation, that the phase is linear in $x$ and $y$ is called the FRAUNHOFER APPROXIMATION. Lec 3

In other words, we can approximate the wavefront as planar at the aperture.

Fortunately, now the argument of the exponential is a linear function of $x$ and $y$.

Integrating from $-\frac{\Delta_x}{2}$ to $\frac{\Delta_x}{2}$ and $-\frac{\Delta_y}{2}$ to $\frac{\Delta_y}{2}$, we have

$$U(P) = Ke^{ik(s_0+r_0)}\frac{2\sin(kX\Delta_x/2)}{kX}\frac{2\sin(kY\Delta_y/2)}{kY}.$$

Now the variable we're interested in is all encoded in $X$ and $Y$. Using $\text{sinc}(x) = \frac{\sin(x)}{x}$, with $\lim_{x\to 0} \text{sinc}(x) = 1$, we have that the peak of the diffraction pattern occurs when $X = 0$ and $Y = 0$. This is the condition for constructive interference.

Geometrically, we can write

$$X = -\sin(\theta_0) + \sin(\theta_p) = 0 \Rightarrow \theta_p = \theta_0.$$

So the peak of the diffraction pattern occurs in the direction of the incident wave. This is a reassuring for us, since this means that light mainly travels in a straight line.

Now, the intensity is proportional to $|U|^2$, so we have

$$I(X) \propto 4\frac{\sin^2(kX\Delta_x/2)}{(kX)^2}\frac{\sin^2(kY\Delta_y/2)}{(kY)^2}.$$

Expanding the fraction and using the definition of sinc, we have

$$I(X) \propto \Delta_x^2\Delta_y^2 \cdot \text{sinc}^2(kX\Delta_x/2) \cdot \text{sinc}^2(kY\Delta_y/2).$$

So the intensity is proportional to the square of the area of the slit.

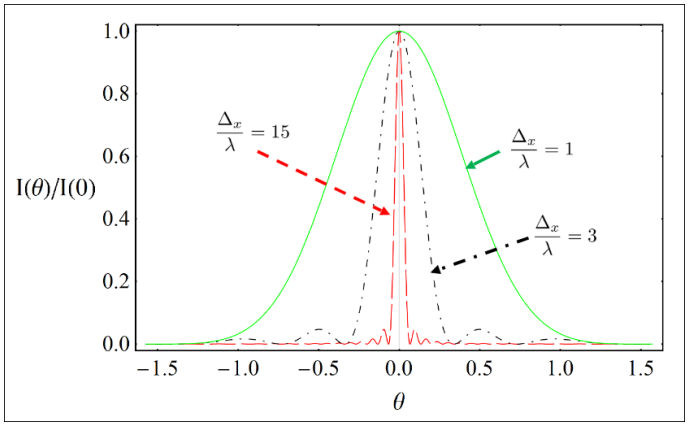



Figure 9: The diffraction pattern of a rectangular slit.

Furthermore, we notice that the sine part oscillates with $X$. The first width of the oscillation can be well characterized by the first zero of the sinc function, which occurs, when $\sin(k\Delta_x X/2) = 0$, or $k\Delta_x X/2 = m \cdot \pi$. Recall, that

$$X = -\sin(\theta_0) + \sin(\theta_p).$$

Simplifying with $\theta_0 = 0$, we have

$$\frac{k\Delta_x}{2}\sin(\theta_p) = m \cdot \pi \Rightarrow \sin(\theta_p) = m\frac{\lambda}{\Delta_x}.$$

The first zero occurs when $m = 1$, so we have

$$\sin(\theta_p) = \frac{\lambda}{\Delta_x}.$$

Suppose now, that we have some filter on the aperture, which is a function $F(x, y) = A(x, y)e^{i\phi(x,y)}$, where $A$ is the amplitude and $\phi$ is the phase shift.

Doing the same game as before, we have

$$U = \sum A_{\text{source}} e^{iks} F(x, y) \cdot e^{ikr}.$$

In the Frauenhofer approximation, we have

$$U(P) \propto \iint F(x, y)e^{ik(Xx+Yy)}\, dx\, dy.$$

Something something Fourier transform...

In an imaging system with a lens before the aperture, we can create planar waves at the aperture. In this case, the Frauenhofer approximation is valid, and we can use the Fourier transform to compute the diffraction pattern. With a second lens, we made an imaging system. With this we can notice that every point turns into some blur.

For a telescope, we wanted to resolve the image of two stars. We now want to know how far do the stars need to be apart in order to be able to be distinguished. This is given by the Rayleigh criterion: If the centre of the image 2 falls on the first zero of image 1, then we can resolve the two stars. This means, that for the angle between the two stars, we have

$$\sin(\theta_s) > \frac{\lambda}{\Delta_x}.$$

The other problem we looked at was the microscope. In this case we have parallel light coming onto an object. We can model this with a transmission function $f(x) = \frac{1}{2}(1 + \cos(k_A x))$. Doing the Frauenhofer approximation, we have

$$\begin{aligned}
&\int_{-\frac{\Delta_x}{2}}^{\frac{\Delta_x}{2}} \frac{1}{2}(1 + \cos(k_A x))e^{-ikXx}\, dx \\
&= \frac{1}{2}\left[\int_{-\frac{\Delta_x}{2}}^{\frac{\Delta_x}{2}} e^{-ikXx}\, dx + \int_{-\frac{\Delta_x}{2}}^{\frac{\Delta_x}{2}} e^{-ikXx+ik_A x}\, dx + \int_{-\frac{\Delta_x}{2}}^{\frac{\Delta_x}{2}} e^{-ikXx-ik_A x}\right. \\
&\propto \frac{\sin(ka\sin(\theta))}{ka\sin(\theta)} + \frac{\sin(ka\sin(\theta) - k_A a)}{ka\sin(\theta) - k_A a} + \frac{\sin(ka\sin(\theta) + k_A a)}{ka\sin(\theta) + k_A a}.
\end{aligned}$$

With this, we find that $\frac{2\pi}{k_a} = \frac{\lambda}{\sin(\theta)}$, where $\sin(\theta)$ is the angle of the light coming from the object. This means, that we can only resolve features of size $> \lambda$, since $\sin(\theta) \leq 1$.

Lec 4

The reason we wanted that $\frac{2\pi}{k_a} = \frac{\lambda}{\sin(\theta)}$ was that we need to catch also the first side maximum of the diffraction pattern. Without this, we would loose the spatial information of the object. This is known as the Abbé limit, which states that the resolution of an optical system is limited by the wavelength of light.

## 1.4 Polarisation of Light

Recall the Wave equation

$$-\nabla^2\mathbf{E} + \frac{1}{c^2}\frac{\partial^2\mathbf{E}}{\partial t^2} = 0.$$

The plane wave solutions are given by

$$\mathbf{E}(\mathbf{r}, t) = \mathbf{E}_0 \cos(\mathbf{k} \cdot \mathbf{r} - \omega t).$$

Since $\nabla \cdot \mathbf{E} = 0$, we find, that $\mathbf{k} \cdot \mathbf{E}_0 = 0$, which means that the electric field is perpendicular to the direction of propagation.

For the magnetic part of the solutions, we get from $\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$, that

$$\mathbf{k} \times \mathbf{E}_0 = \omega \mathbf{B}_0.$$

Hence,

$$\mathbf{B}(\mathbf{r}, t) = \mathbf{B}_0 \cos(\mathbf{k} \cdot \mathbf{r} - \omega t).$$

With this, we note, that we can describe polarised light through the direction of the *electric* field. By linearity of the wave equation, we can also have superpositions of plane waves. For example:

$$\mathbf{E}(\mathbf{r}, t) = E_x \hat{x} \cos(\mathbf{k} \cdot \mathbf{r} - \omega t) + E_y \hat{y} \cos(\mathbf{k} \cdot \mathbf{r} - \omega t + \alpha).$$

If $\alpha = 0$, we notice that the polarisation moves along a line, hence this is called LINEAR POLARISATION. If $E_x = E_y$ and $\alpha = \pm\frac{\pi}{2}$, we notice that the polarisation moves along a circle, hence this is called CIRCULAR POLARISATION.

If $\alpha = -\frac{\pi}{2}$ we have lefthand circular polarisation, and if $\alpha = \frac{\pi}{2}$ we have righthand circular polarisation.

Since many light sources have atoms with randomized directions, the light is often unpolarised.

When we have a surface, we will often times have a well defined direction. Given a surface, we can define the normal vector. If the normal vector lies in the same plane as spanned by $\mathbf{k}$ and $\mathbf{E}$, we call this P-POLARISATION for PARALLELE POLARISATION.

The second case is when the normal vector is in the plane spanned by $\mathbf{k}$ and $\mathbf{B}$. This is called S-POLARISATION for SENKRECHTE POLARISATION. Usually p-polarisation is depicted with a red arrow, and s-polarisation is depicted with a red dot.

Let us quickly recall linear dielectric media. If there is no electric field, then $\mathbf{D} = 0$. If we apply an electric field, the material will polarise, i.e. produce a bound charge. The relationship between the electric field and the displacement field is given by

$$\mathbf{D} - \mathbf{P} = \epsilon_0 \mathbf{E}.$$

Using, that in linear media, $\mathbf{P} = \varepsilon_0 \chi_e \mathbf{E}$, we find

$$\mathbf{D} = \epsilon_0(1 + \chi_e)\mathbf{E} = \epsilon_0 \epsilon_r \mathbf{E}.$$

Here $\chi_e$ is the electric susceptibility, and $\epsilon_r$ is the relative permittivity. The magnetic field is similar, and we have

$$\mathbf{B} = \mu_0(1 + \chi_m)\mathbf{H} = \mu_0 \mu_r \mathbf{H}.$$

In linear materials, Maxwells equations are given by

$$\nabla \cdot \mathbf{D} = 0, \quad \nabla \cdot \mathbf{B} = 0, \quad \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad \nabla \times \mathbf{H} = \frac{\partial \mathbf{D}}{\partial t}.$$

With this, we find that the wave equation in linear media is given by

$$-\nabla^2 \mathbf{E} + \mu_0 \mu_r \epsilon_0 \epsilon_r \frac{\partial^2 \mathbf{E}}{\partial t^2} = 0.$$

Hence, the speed of light in a linear medium is given by

$$v = \frac{1}{\sqrt{\mu_0 \mu_r \epsilon_0 \epsilon_r}} = \frac{c}{n}.$$

Now, also

$$\mathbf{B}_0 = \frac{n \mathbf{k}_0 \times \mathbf{E}_0}{\omega} \quad |\mathbf{B}_0| = \frac{n |\mathbf{E}_0|}{c}.$$

When a electric wave hits a surface, we have to consider the boundary conditions.

**Theorem 1.5: Boundary Conditions for Electromagnetic Waves**

The boundary conditions for electromagnetic waves are given by

$$B_1^\perp = B_2^\perp, \qquad H_1^\parallel = H_2^\parallel,$$
$$D_1^\perp = D_2^\perp, \qquad E_1^\parallel = E_2^\parallel.$$

**Proof.** To apply the boundary conditions, we take mathematical relations and apply them to the surface. For example, by the divergence theorem, we have

$$\iiint_{\mathcal{V}} \nabla \cdot \mathbf{B} \mathrm{d}\tau = \oiint_{\partial \mathcal{V}} \mathbf{B} \cdot \mathrm{d}\mathbf{a}.$$

From Maxwells equations, we know that $\nabla \cdot \mathbf{B} = 0$. So

$$\oiint_{\partial \mathcal{V}} \mathbf{B} \cdot \mathrm{d}\mathbf{a} = 0.$$

Now, consider a pillbox with the top and bottom surface parallel to the surface, of thickness $\delta$, where half the surface is in the medium, and the other half is on the outside.

Letting $\delta \to 0$, we find that

$$B_1^\perp = B_2^\perp.$$

The other boundary condition is derived from Stokes Theorem, which states that

$$\iint_{\mathcal{S}} \nabla \times \mathbf{H} \mathrm{d}\mathbf{a} = \oint_{\partial \mathcal{S}} \mathbf{H} \cdot \mathrm{d}\mathbf{l}.$$

From Maxwells equations, we know that $\nabla \times \mathbf{H} = \frac{\partial \mathbf{D}}{\partial t}$, so

$$\iint_{\mathcal{S}} \frac{\partial \mathbf{D}}{\partial t} \mathrm{d}\mathbf{a} = \oint_{\partial \mathcal{S}} \mathbf{H} \cdot \mathrm{d}\mathbf{l}.$$

Now given a boundary, we can take a rectangular loop with two sides parallel to the surface, and two sides perpendicular to the surface. Letting the length of the perpendicular sides go to zero, notice, that the integral on the left hand side goes to zero, since the area goes to zero. From the right hand side, we find that

$$H_1^\parallel = H_2^\parallel.$$

An analogous argument can be made for the electric field, and we find that

$$D_1^\perp = D_2^\perp, \quad E_1^\parallel = E_2^\parallel.$$

□

In the following, we will focus on di-electrics with no magnetic component. So $\varepsilon_1 \neq \varepsilon_2$, but $\mu_1 = \mu_2$.

Suppose we have a p-polarised wave incident on a plane. We call the incident wave $\mathbf{E}_i$, the reflected wave $\mathbf{E}_r$, and the transmitted wave $\mathbf{E}_t$. The incident angle to the normal is $\theta_i$, the reflected angle is $\theta_r$, and the transmitted angle is $\theta_t$. In the following we will assume Snell's law, however it could be derived from the boundary conditions of the magnetic field. We know that

$$\theta_r = \theta_i, \quad n_1 \sin(\theta_i) = n_2 \sin(\theta_t),$$

where $n_1 = \sqrt{\mu_1 \varepsilon_1}$ and $n_2 = \sqrt{\mu_2 \varepsilon_2}$ are the refractive indices of the two media. Since the materials are non-magnetic, we have $\mu_1 = \mu_2 = 1$, so $n_1 = \sqrt{\varepsilon_1}$ and $n_2 = \sqrt{\varepsilon_2}$.

From $H_1^\parallel = H_2^\parallel$, we find that using $\mu_1 = \mu_2 = 1$,

$$B_i + B_r = B_t.$$

Since all point in the same direction, and using $|\mathbf{B}_0| = \frac{n|\mathbf{E}_0|}{c}$, we find that

$$n_1(E_i + E_r) = n_2 E_t.$$

From $E_1^{\parallel} = E_2^{\parallel}$, we find that

$$E_i \cos(\theta_i) - E_r \cos(\theta_r) = E_t \cos(\theta_t).$$

Using $\theta_r = \theta_i$, we find that

$$(E_i - E_r)\cos(\theta_i) = E_t \cos(\theta_t).$$

Rearranging, we find that

$$\begin{aligned}
\frac{n_1}{n_2}(E_i + E_r) &= \frac{E_i - E_r}{\cos(\theta_i)} \cos(\theta_t) \\
\frac{n_1}{n_2}\left(1 + \frac{E_r}{E_i}\right) &= (1 - \frac{E_r}{E_i})\frac{\cos(\theta_t)}{\cos(\theta_i)} \\
\frac{E_r}{E_i}\left(\frac{n_1}{n_2} + \frac{\cos(\theta_t)}{\cos(\theta_i)}\right) &= \frac{\cos(\theta_t)}{\cos(\theta_i)} - \frac{n_1}{n_2}
\end{aligned}$$

.

Hence,

$$\frac{E_r}{E_i} = \frac{n_2 \cos(\theta_i) - n_1 \cos(\theta_t)}{n_2 \cos(\theta_i) + n_1 \cos(\theta_t)}.$$

Doing some trigonometry, we can find that

$$\frac{E_r}{E_i} = \frac{\tan(\theta_i - \theta_t)}{\tan(\theta_i + \theta_t)}.$$

This is zero, when $\theta_i + \theta_t = \frac{\pi}{2}$. Then no light gets reflected. Using Snell's law, we find that this happens when

$$\theta_i = \arctan\left(\frac{n_2}{n_1}\right).$$

This is known as the BREWSTER ANGLE.

This effect is for example useful when taking holidays at the sea. The light reflected from the water is mainly s-polarised, so if you wear polarised sunglasses, you can reduce the glare from the water.

Lec 5

## 1.5 Spectroscopy

We now want to discuss how different colors of light can be used and how we can separate colors. One device to do that is the prism. This is the idea behind spectroscopy. In general, spectroscopy relies on separating colours/wavelength/frequency into different spatial features.

To achieve this we want to use interference. For this we split up the light and send it down different paths. In the prism, the light is split up into different colors because of the different refractive indices for different wavelengths. By Snell's law, we know, that the angle of refraction then changes with the wavelength.

The first interference spectrometer we look at is a GRATING SPECTROMETER. For this, we have a source which sends out some noisy light. We use a lens to focus the light into a slit. This is used to limit, where light comes from. In addition we have a focal lens a distance $f$ away from the slit. This will produce planar waves after slit. In the path of the light, we put a GRATING. It is tilted at an angle $\theta_i$ to the optical axis. Of the grating, there will be some light reflected at an angle $\theta_i$. Afterwards, we need another lens to focus the light onto a given point on the screen. The question is now, at which angles we see interference and at which angles we see destructive interference.

We can also implement a reference, with a known source. Then we can compare the interference pattern of the known source with the unknown source.

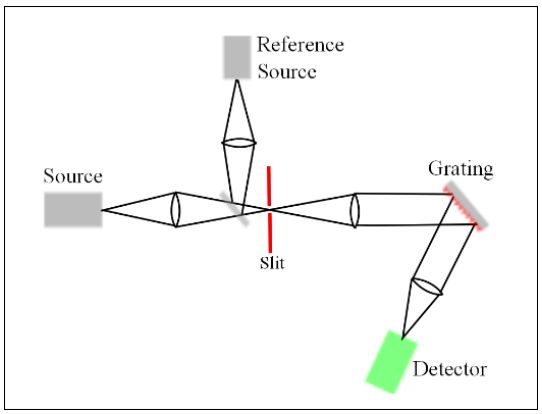



Figure 10: Grating Spectrometer

The grating is a surface with places that can reflect light and places that cannot. In general, we want to have a periodic structure. Since light is a plane wave, we can assume, that all light is coming from the same angle. By the wavelike nature of light, we can have a outgoing angle of $\theta_j$ which is not nesessarily equal to the incoming angle $\theta_i$.

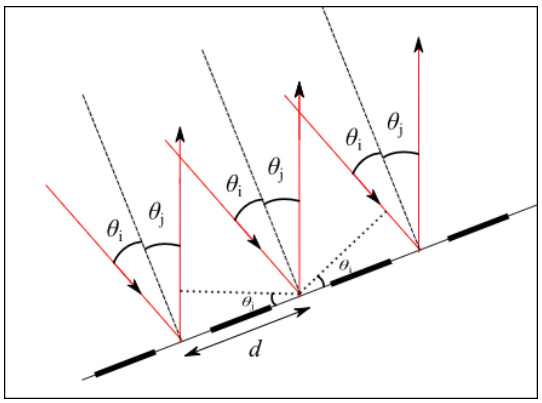



Figure 11: Grating

We now want to find the phase difference of the outgoing light. The phase shift is given by $\delta\phi = k\Delta x$, where $k$ is the wave number and $\Delta x$ is the path difference.

The extra distance the has to travel between slit $n$ and $n+1$ is given by

$$\Delta x_{\text{incoming}} = d\sin(\theta_i) \quad \text{and} \quad \Delta x_{\text{outgoing}} = d\sin(\theta_j).$$

So in total, $\Delta x = d(\sin(\theta_j) - \sin(\theta_i))$. The phase difference is then given by

$$\delta\phi = kd(\sin(\theta_j) - \sin(\theta_i)).$$

To see light under the angle $\theta_j$, we need to have constructive interference. This happens, that $\delta\phi = 2\pi\mathbb{Z}$.

Over the full grating, we have an initial amplitude $A(0)$, and we sum up over all the slits.

$$\begin{aligned}A(\theta_j) &= A(0) + A(0)e^{i\delta\phi} + A(0)e^{2i\delta\phi} + \cdots + A(0)e^{i(N-1)\delta\phi} \\ &= A(0)\sum_{n=0}^{N-1} e^{in\delta\phi} \\ &= A(0)\frac{1-e^{iN\delta\phi}}{1-e^{i\delta\phi}} \\ &= A(0)\frac{e^{iN\delta\phi/2}-e^{-iN\delta\phi/2}}{e^{i\delta\phi/2}-e^{-i\delta\phi/2}}e^{i(N-1)\delta\phi/2} \\ &= A(0)\frac{\sin(N\delta\phi/2)}{\sin(\delta\phi/2)}e^{i(N-1)\delta\phi/2}.\end{aligned}$$

Thus, for the intensity we have

$$I(\theta_j) = |A(\theta_j)|^2 = |A(0)|^2\frac{\sin^2(N\delta\phi/2)}{\sin^2(\delta\phi/2)}.$$

When the denominator goes to zero, we get peaks of height $N^2$. All other features have height $\approx 1$. Notice, that since the period of constructive interference is dependent on the wavelength, we can use this to separate different wavelengths. We now want to find a condition like the Rayleigh criterion, but for spectral resolution. I.e. when does the peak of one color lie on the first zero of the other color.

For this, we can write

$$\delta\phi = 2\pi\frac{d}{\lambda}(\sin(\theta_j) - \sin(\theta_i)).$$

For peaks, we can have seen that $\delta\phi = 2\pi m$, where $m$ is an integer. Note, that at these points, the numerator is also zero. So the limit of the function is finite. Note, that the numerator is zero if $k\pi = N\delta\phi/2$, where $k$ is an integer. Since we have a peak if $k = Nm$, the nearest zero is at $k = Nm + 1$. Thus, the nearest zero is at

$$\frac{N\delta\phi}{2} = (Nm+1)\pi.$$

The condition we want is that when we have $\lambda_1, f_1$ constructively interfering at an angle $\theta_j$, we need to get destructive interference for $\lambda_2, f_2$ at the same angle. This means, that we need

$$\begin{aligned}\delta\phi_1 &= k_1 d(\sin(\theta_j) - \sin(\theta_i)) = 2\pi m \\ \delta\phi_2 &= k_2 d(\sin(\theta_j) - \sin(\theta_i)) = 2\pi m + \frac{2\pi}{N}.\end{aligned}$$

Combining and rearranging, we get

$$k_2 - k_1 = \frac{2\pi}{Nd(\sin(\theta_j) - \sin(\theta_i))}.$$

In terms of frequencies, we have $k = 2\pi f/c$. Thus, we can write

$$\delta f = \frac{c}{Nd(\sin(\theta_j) - \sin(\theta_i))}.$$

Notice, that $Nd$ corresponds to the total length of the grating $w$. Since sine is bounded by 1, we find the maximal value of $Nd(\sin(\theta_j) - \sin(\theta_i))$ is $2w$. Thus, the smallest frequency difference we can resolve is

$$\delta f_{\min} = \frac{c}{2w}.$$

This corresponds to the time it takes light to travel the longest path difference we can introduce.

To make a long path difference, we can for example use a Michelson Spectrometer. Again, we have a source, a slit and a lens. The light is the split by a beam splitter. At the ends of both paths, there are mirrors. The light is reflected back and recombined at the beam splitter. The light is then focused onto a screen.

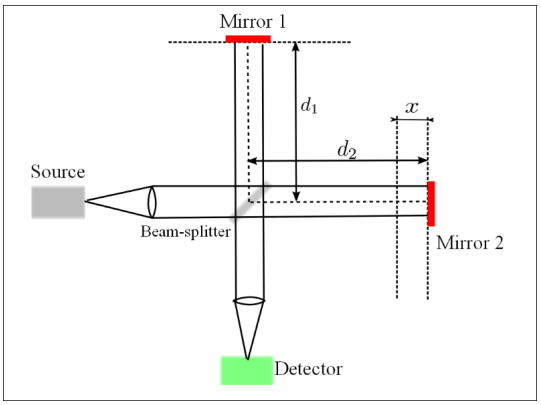



Figure 12: Michelson Spectrometer

The difference in phase for light travelling via mirror 2 versus mirror 1 is given by

$$\delta\phi = k\Delta x = 2k(d_2 - d_1).$$

Using a complex wave at the detection, we can find a wave of the form

$$A = \frac{A_0}{2}\left(1 + e^{i\delta\phi}\right).$$

Defining $x \coloneqq d_2 - d_1$, we can write

$$I(x) = \frac{|A|^2}{|A_0|^2} = \frac{1}{2}|1 + \cos(\delta\phi)| = \frac{1}{2}(1 + \cos(4\pi x\frac{f}{c})).$$

We can rewrite the amplitude of the original light as seen as

$$I(x) = \frac{A_0^2}{2}(1 + \cos(\delta x)) = A_0^2\cos^2(\delta x/2).$$

This is one, if $\delta x = \pi n$ and zero if $\delta x = \pi(n + 1/2)$. The distance between two zeros is then $\Delta x = \pi$.

So for dark spots, we require

$$2x = (n + \frac{1}{2})\lambda.$$

For bright spots, we require

$$2x = n\lambda.$$

Performing the experiment, we expect to see either a bright or a dark but fringe. However, we can see stripes. This implies that one mirror is tilted. If the mirror was perfectly aligned, we should see rings, if the beam is not perfectly collimated.

We think about power density in a frequency range from $f$ to $f+\delta f$. The power density between $f$ and $f+\delta f$ is given by $S(f)\delta f$. We claim that the intensity in a Michelson

spectrometer is given by as a sum of contributions from all frequencies. Thus,

$$I_{\text{tot}}(x) = \text{const} \cdot \sum \text{all} I(x) \text{from diffrent} f.$$

Letting $\delta f \to 0$, we can write this as an integral

$$I_{\text{tot}}(x) = \text{const} \cdot \int S(f) I(x) df \propto \int S(f) \frac{1}{2}(1+\cos(4\pi x \frac{f}{c})) df.$$

Expaning, we get

$$I_{\text{tot}}(x) = I_0 + \frac{1}{2}\int_0^\infty S(f) \mathrm{d}f \cos(4\pi x \frac{f}{c}).$$

So $(2I_{\text{tot}}(x) - I_0)$ is the Fourier transform of $S(f)$. Inverting the fourier transform, we get

$$S(f) = \int_0^{x_{\max}} (2I_{\text{tot}}(x) - I_0) \cos(4\pi x \frac{f}{c}) dx.$$

Lec 6

Note, that in the sum the different frequencies will have cross terms, however since we measure over a long time scale, these will average to zero. Furthermore, we can only integrate to $x_{\max}$, since we cannot measure from 0 to $\infty$. Thus, we can write

$$\begin{aligned} S_{\text{meas}}(f_1) &= \int_0^{x_{\max}} dx\, (2I_0 - I_0) \cos(4\pi x \frac{f_1}{c}) \\ &\propto \int_0^\infty \mathrm{d}f\, S(f) \int_0^{x_{\max}} dx\, \cos(4\pi x \frac{f}{c}) \cos(4\pi x \frac{f_1}{c}). \end{aligned}$$

Using the identity $\cos(a)\cos(b) = \frac{1}{2}(\cos(a-b)+\cos(a+b))$, we can write

$$\begin{aligned} S_{\text{meas}}(f_1) &\propto \int_0^\infty \mathrm{d}f\, S(f) \int_0^{x_{\max}} dx\, \frac{1}{2}\left(\cos(4\pi x \frac{f-f_1}{c}) + \cos(4\pi x \frac{f+f_1}{c})\right) \\ &\propto \int_0^\infty \mathrm{d}f\, S(f) \left(\frac{\sin(2\pi x_{\max}\frac{f-f_1}{c})}{2\pi x_{\max}\frac{f-f_1}{c}} + \frac{\sin(2\pi x_{\max}\frac{f+f_1}{c})}{2\pi x_{\max}\frac{f+f_1}{c}}\right) \\ &= \int_0^\infty \mathrm{d}f\, S(f) \left(\text{sinc}(2\pi x_{\max}\frac{f-f_1}{c}) + \text{sinc}(2\pi x_{\max}\frac{f+f_1}{c})\right) \end{aligned}$$

Assuming that we started with a single frequency, $S(f) = I_0\delta(f-f_0)$, where $f_0$ is the source frequency, and $f_1$ is the frequency we are measuring. For $f_0 \approx f_1$, we can ignore the second term, and we get

$$S_{\text{meas}}(f_1) \propto I_0 \left(\text{sinc}(2\pi x_{\max}\frac{f_0-f_1}{c})\right).$$

Because of the Rayleigh criterion, we will say that we describe the broadness of the peak by the first zero of the sinc function. This is located at

$$\frac{4\pi}{c} x_{\max} \underbrace{(f_0 - f_1)}_{\delta f} = \pi \Rightarrow \delta f = \frac{c}{4x_{\max}}.$$

Note, that $4x_{\max}$ is two times the maximum path difference, where the factor two appears because the light travels to the mirror and back. Thus, we can write

$$\frac{\delta f}{f} = \frac{\lambda}{4x_{\max}}.$$

In conclusion, the resolution is set by the longest path difference one can create.

Suppose now, we have a source with two different frequencies $f_1$ and $f_2 = f_1 + \delta f$. The total intensity is now

$$I_{\text{tot}}(x) = \frac{1}{2}(1 + \cos(2\pi x \delta f/c)\cos(2\pi x(2f_1+\delta f)/c)).$$

The second term produces far fringes. The first term moves rather small compared, so we call it the ENVELOPE. The fast fringes looks like a normal single frequency at frequency $2f_1 + \delta f$. The envelope has a period of $c/\delta f$. In order to see we have two different frequencies, we need to see the change in contrast from its maximum to its minimum. For this, we need at least half a period of the envelope. Thus, we need to have a path difference of at least

$$D = \frac{2\pi x}{c}\delta f = \frac{\pi}{2} \Rightarrow \delta f = \frac{c}{4\delta x}.$$

Since $D = 2x$, we again arrive at

$$\delta f = \frac{\lambda}{2D}.$$

To get a long path difference, on limited space, we can use a FABRY-PEROT ETALON. A etalon is essentiall 2 flat parallel reflective surfaces at a distance $d$. Strictly speaking, the etalon has a fixed distance.

Let's determine what happens to the phase of light when it is reflected back and forth between the two surfaces. The incoming wave has the form $E_{\text{in}} = A_0 e^{i(kx-\omega t)}$. We will assume that the mirrors are the same. Then, light can be transmitted $T = t^2$ $E_{\text{in}} \to tE_{\text{in}}$. If we instead reflex, $R = |e^{i\phi}r|^2$, $E_{\text{in}} \to re^{i\phi}E_{\text{in}}$. Lastly light can travel a distance $d$, $E_in \to e^{ikd}E_{\text{in}}$. Let us just consider the light that is transmitted through the etalon. The first transmitted light is

$$A_0' = A_0 t e^{ikd} t = A_0 t^2 e^{ikd}.$$

What if we reflect twice, then we have

$$A_1' = A_0 t e^{ikd} r e^{i\phi} e^{ikd} r e^{i\phi} t = A_0 t^2 r^2 e^{ikd} e^{i2(kd+\phi)}.$$

So we get

$$\frac{A_1'}{A_0'} = Re^{i2(kd+\phi)}.$$

So

$$An' = A_0 t^2 e^{ikd}(Re^{i\delta\phi})^n.$$

So for the output power, we get

$$A_{\text{out}} = A_0 t^2 e^{ikd} \sum_{n=0}^\infty (Re^{i\delta\phi})^n = A_0 t^2 e^{ikd} \frac{1}{1 - Re^{i\delta\phi}}.$$

So we also find

$$\begin{aligned} \frac{I}{I_0} &= \frac{T^2}{1 + R^2 - 2R\cos(\delta\phi)} \\ &= \frac{(1-R)^2}{(1-R)^2 + 2R(1-\cos(\delta\phi))} \\ &= \frac{1}{1 + \frac{2R}{(1-R)^2}(1-\cos(\delta\phi))} \\ &= \frac{1}{1 + \frac{4R}{(1-R)^2}\sin^2(\delta\phi/2)}. \end{aligned}$$

This tells us, that the maxima of this function are, when $\delta\phi = 2\pi\mathbb{Z}$. For $R \to 1$, we see that the peaks get sharper and sharper. The peak spacing is given by $\delta\phi = 2\pi$.

We can now combine a Fabry-Perot etalon with a diffraction grating. The etalon will produce sharp peaks, and the diffraction grating will separate the different wavelengths. Lets focus on the etalon first. After the etalon, notice, that the peak condition is

$$2\pi n = 2kd + 2\phi.$$

Since $2\phi$ is a constant, we see that we get a train of peaks, one per order if

$$\Delta d = \frac{\pi}{k} = \frac{\lambda}{2}.$$

This also explains why we need a grating. It serves as a large scale filter, so that we can separate the different orders.

So if we have two frequencies. For different $k$, we will need different $d$.

So once again, given two frequencies, how far can we measure? The limit is given, if one order falls exactly where the second order is of the other frequency. This is given by

$$2(k_2 - k_1)d = 2\pi \Rightarrow \Delta f = \frac{c}{2d}.$$

This quantity is called the Free spectral range. It describes the maximum frequency difference, that can be measured with a Fabry-Perot etalon. Going back to the airy function, we define the Full width half maximum (FWHM) as the width of the peak at half its maximum.

Lec 7

The free spectral range tells us how far apart the resonant frequencies of a cavity are.

We now want to determine how many frequencies we can tell apart. For this, let us look at the airy function again. For half the intensity, we require

$$1 = \frac{4R}{(1-R)^2} \sin^2\left(\frac{\delta}{2}\right).$$

Doing a small angle approximation, we have

$$1 = \frac{4R}{(1-R)^2} \left(\frac{2\pi\delta f}{c} d\right)^2.$$

So

$$\delta f = \frac{c}{2d} \frac{1-R}{2\pi\sqrt{R}}.$$

Since this is only one side of the peak, we have to multiply by 2 to get the FWHM:

$$\delta f_{FWHM} = \frac{c}{2d} \frac{1-R}{\pi\sqrt{R}} = FSR \underbrace{\frac{1-R}{\pi\sqrt{R}}}_{\frac{1}{F}}.$$

Where we define the Finesse as

$$F = \frac{FSR}{\delta f_{FWHM}} = \frac{\pi\sqrt{R}}{1-R}.$$

Note, that the Finesse only depends on the reflectivity of the mirrors, and not on the length of the cavity. The Finesse tells us how many peaks we can resolve in a given frequency range.

The Finesse tells us how many peaks we can resolve in a given frequency range. It is a measure of the longest path in the etalon, which can be estimated by $d \cdot F$.

Experimentally, we can see that a Fabri-Perot cavity has rings as an interference pattern. This is why we need a slit to select a single mode. The intuition is that the round trip path length depends on the incoming beam angle. This is because one round trip actually takes a distance $\frac{2d}{\cos\theta}$, where $\theta$ is the angle of incidence. Furthermore, there is an extra path length of

$$d'' = 2d \tan\theta \sin\theta.$$

The rings are bright, if $2d\cos(\theta) = m\lambda$. With increasing $\theta$, the rings get closer together.

# 2 Photons

## 2.1 Photoelectric effect

Consider the following setup. A mercury lamp passes through a filter and hits a zinc plate, which is connected to an electrometer, measuring static charge. If there is no charge, we excpect no change. If we have negative charges on the electrometer, we see, that the plate gets discharged when light shines on it. However, using a glass plate as a filter, no discharge occurs. If the plate was instead of quartz, the discharge occurs. The difference between them is, that glass has a transmission curve between 300 and 2000 nm, while quartz has a transmission curve between 140 and 4000 nm. So the effect has to do with UV or IR light.

Using $I \propto |\mathbf{E}\cos(kz-\omega t)|^2$, we would excpect the electrons in the plate to move with

$$x = \frac{eE_0}{m\omega^2}\cos(kz-\omega t) \Rightarrow v = -\frac{eE_0}{m\omega}\sin(kz-\omega t).$$

Hence,

$$\langle E_{kin}\rangle = \frac{1}{2}m\langle v^2\rangle = \frac{e^2E_0^2}{4m\omega^2}.$$

This grows with intesity, but also, higher frequencies imply lower kinetic energy. But this is *not* what we observe. But clearly, all of this is wrong, because the mercury lamp has an emission along 254nm to 600nm. So something is wrong with our classical picture.

This is described in the PLANCK EXPERIMENT. For this we have a mercury lamp, a selective frequency filter and a detector. The detector here is a PHOTOTUBE. For this, we have a amperemeter connected to a cathode and an anode. We also put this setup in a vacuum. We also have a variable voltage source, which can be used to apply a voltage between the cathode and the anode. Let $U = V_A - V_C$. If this is less than 0, it slows down the electrons, if its bigger, it accelerates them. If this works, the electrons leave the cathode with $E_{\text{kin}}$. It then arrives at the anode if $E'_{\text{kin}} \geq 0$ where $E'_{\text{kin}}$ is the energy at the anode. From energy conservation, we have

$$E_{\text{kin}} - eV_c \stackrel{!}{=} E'_{\text{kin}} \geq 0 \Rightarrow E_{\text{kin}} \geq e|V_c|.$$

Hence there should be some breaking voltage $U_B$ where $e|U_B| > E_{\text{kin}}$.

Measuring the breaking voltage for different wavelengths, we find that the braking voltage is linear. In the following we will use $\nu$ for the frequency of the light. The slope of the linear fit is given by

$$\frac{\Delta U_B}{\Delta \nu} \approx 4 \times 10^{-15}\,\text{V s}..$$

Hence,

$$\frac{\Delta E_{\text{kin}}}{\Delta \nu} = 6.57 \times 10^{-34}\,\text{J s} \approx h.$$

Where $h$ is the PLANCK CONSTANT.

If we instead measure the intensity of the current, we find the shape of an S-curve.

Max Planck in 1900 sugessted, that electromagnetic waves can only have "packets" of energy. 5 years later, Einstein suggested, that in each packet, called QUANTA, there is an energy $E = h\nu$. This quanta later was called a PHOTON. Now how does this explain our observations? For the breaking voltage depending linear on frequency, not intensity, we can see that if one photon is absorbed by one electron, then $h\nu = E_{\text{kin}} + const.$ Denoting the constant as $h\nu_{\text{min}}$, we have

$$E^B_{\text{kin}} = h\nu - h\nu_{\text{min}}.$$

If we look at the limiting case, we find

$$E^B_{\text{kin}} = h\nu_{\text{min}} = eV_W.$$

Here $W$ is called the WORK FUNCTION. It describes the minimal energy needed to remove an electron from a solid. The work function depends on the material and surface properties.

For the observed S-curve, the power is the photon energy times the flux

$$P = h\nu \cdot \dot{N}p\nu \Rightarrow I_S = e\dot{N}p\nu = \frac{eP}{h\nu}.$$

This fixes the saturation and the shape of the S-curve is a material property.

As an application, Photomultiplier tubes (PMT) are used to detect single photons. For this, after the cathode, we place DYNODES. Theses are charge multipliers via acceleration. From one dynode to the next, there is a voltage difference of about 100 V. So 1 electron can knock out several electrons from the next dynode. This is useful since the current of one electron is too small to be measured. The PMT can have 10-12 dynodes, so the current can be amplified by a large factor. Such PMTs are for example used in ICEcube, which won the nobel prize 2 days ago.

So what about the inverse process? Suppose a free electron with kinetic energy $E_{\text{kin}}$ hits a metal surface. The kinetic energy is then converted into a photon.

For visible light, like $\lambda = 500$ nm, we require $h\nu = 2.5$ eV. For x-rays, with $\lambda = 0.1$ nm, we require $h\nu = 12$ keV. This is doable, but quite hard. For something even more energetic like gamma rays with $\lambda = 1pm$, we require $h\nu = 1$ MeV. This is too hard to achieve with electrons.

This is indeed, how we can produce x-rays. For this we have electrons accelerated through a vauum tube, which then hit a metal target (anode). For the electron source we can use a heated FILAMENT. If it's hot enough for electrons to overcome the work function, they can leave the filament. This is called THERMIONIC EMISSION. After the electrons get accelerated, they hit the anode. This deceleration creates radiation. In reality, about 1% of the energy is converted into x-rays, the rest is heat. For this we need to water cool the anode. The emission is

$$h\nu_{\text{max}} = eU + W \approx eU.$$

From this, we get

$$\lambda_{\text{min}} = \frac{hc}{eU}.$$

Looking at the emission spectrum, we notice a weird feature for high energies. This is due to interference for x-rays. For visible light, we had $d\sin\theta = m\lambda$. Since we

needed $\sin\theta \leq 1$, we needed $\lambda$ and $d$ in the same size. But now, if $\lambda$ is 0.1 nm, we need $d$ to be of the same size. This is quite hard to produce. We can replace our slits by atoms, so we are talking about a crystal. In order to have coherent light, we need to have elastic scattering, i.e. the electrons should not lose energy.

Lec 8

Let us look how the X-rays can interact with the bound electrons in the atoms. For this, we can approximate the binding potential of the electrons in the atom as a harmonic oscillator potential. As such, we can characterize the electron as a mass on a spring with resonance frequency $\omega_0$. The X-rays then drive this mass. We get the equation of motion

$$m\ddot{x} + m\omega_0^2 x = -eE\cos(\omega t).$$

Hence,

$$x_0 = \frac{eE}{m(\omega^2 - \omega_0^2)}.$$

In general, we assume $\omega > \omega_0$, so $x_0 > 0$. This corresponds to a dipole, which radiates at driving frequency $\omega$ with fixed phase. Since we did some approximations, this result is only valid if $x_0$ stays in the harmonic region and the incoming field should not go too close to the resonance frequency $\omega_0$. This model is called Bragg scattering.

Recall that for a diffraction grating, we have the condition for constructive interference

$$n\lambda = d\sin(\theta).$$

For Bragg scattering, the interference condition is given by

$$2a\sin(\theta) = n\lambda.$$

Where $a$ is the distance between the planes of atoms in the crystal and $\theta$ is the angle of incidence. To detect the signal, we place the detector at the angle $2\theta$. In that sense, the crystal acts like a mirror.

However, a crystal has different planes. This allows us to have even more interference conditions.

Until now, we have considered the photon as elastic. Now we will consider it as inelastic. This is called Compton scattering. For this we want to give the photon a momentum. For this, we use the energy-momentum relation for the photon

$$E^2 = p^2c^2 + m^2c^4.$$

Since the photon is massless, we have $E = pc$. Hence, the momentum of the photon is

$$p = \frac{E}{c} = \frac{h\nu}{c} = \frac{h}{\lambda}.$$

Since the photon is inelastic, we want to behave the electron like a free particle. Hence, $h\nu$ has to be bigger than the binding energy of the electron. To make this work, we typically need $\gamma$-rays. The typical experiment to look at this is to have a source like $^{137}$Cs, which emits $\gamma$-rays with energy $E = 662$ keV. The rays then hit a copper rod, which scatters the rays. The rays then hit on a NaI detector, after which a photomultiplier is placed. We use NaI because Iodine is heavy which stops the gamma rays and then reemitts them as visible, by the PMT detectable light.

To determine the scattering, consider a photon with energy $E = h\nu$ hitting an electron at rest. The photon is scattered at an angle $\theta$ with energy $E' = h\nu'$.

Energy conservation gives us

$$h\nu + m_e c^2 = h\nu' + \sqrt{p_e^2c^2 + m_e^2c^4}.$$

Meanwhile from momentum conservation, we have

$$\begin{aligned}\frac{h\nu}{c} &= \frac{h\nu'}{c}\cos(\theta) + p_e\cos(\phi)\\ 0 &= \frac{h\nu'}{c}\sin(\theta) - p_e\sin(\phi).\end{aligned}$$

Squaring and adding the two momentum equations, we get

$$h^2\nu^2 + h^2\nu'^2 - 2h^2\nu\nu'\cos(\theta) = p_e^2c^2.$$

Written more compactly, we have

$$p_e^2c^2 = h^2\nu^2 + h^2\nu'^2 - 2h^2\nu\nu'\cos(\theta).$$

And

$$p_e^2c^2 + m_e^2c^4 = [h(\nu - \nu') + m_ec^2]^2.$$

Together, this gives

$$\begin{aligned}-2h^2\nu\nu'\cos(\theta) &= -2h^2\nu\nu' + 2h(\nu-\nu')m_ec^2\\ h^2\nu\nu'(1-\cos(\theta)) &= h(\nu-\nu')m_ec^2\\ \frac{h}{m_ec^2}(1-\cos(\theta)) &= \frac{1}{\nu'} - \frac{1}{\nu} = \frac{\lambda'-\lambda}{c}.\end{aligned}$$

We usually call $\lambda_C = \frac{h}{m_ec}$ the Compton wavelength. It describes the equivalent wavelength of an electron. Hence, we have

$$\lambda' - \lambda = \lambda_C(1-\cos(\theta)).$$

In terms of frequency, we have

$$\frac{\nu'}{\nu} = \frac{1}{1 + \frac{\nu\lambda_C}{c}(1-\cos(\theta))}.$$

The compton wavelength is $\lambda_C = 2.43\times10^{-12}$ m. At 90 degrees, we see that $\delta\lambda = \lambda_C$. This also, what compton measured in his original experiment.

In conclusion, we see that $\delta\lambda$ is maximal for back scattering, i.e. $\theta = 180°$. Furthermore, $\delta\lambda$ is independent of the incoming wavelength $\lambda$. With gamma rays, at 90 degrees, we see that more than 50% of the energy is transferred.

# 3 Atoms and their Composition

Let us look at the periodic table. In it, the atoms are ordered by their atomic number $Z$. We denote atoms with the notation ${}^{A}_{Z}X$ where $A$ is the mass number, $Z$ is the atomic number and $X$ is the chemical symbol. To find relative masses, historically, the ideal gas law $PV = Nk_BT$ was used, which tells that every gas contains the same number of particles.

One can then fill up two containers with different gases at the same temperature and pressure. Then, one can measure the mass of the two containers. This gives a ratio of the masses. So to get absolute masses, we only need one as a reference. The idea is to take a perfect crystal, then measure the latice spacing ($a$) with X-ray diffraction. Then

$$m_{\text{atom}} = \frac{a^3\rho}{n_{\text{cell}}},$$

where $\rho = m/V$ is the density. Up to 2019 this was done by comparing it to a mass standard. Since 2019, the mass is defined by the Planck constant $h$.

A better way to determine masses is to do mass spectrometry. This is useful, since different isotopes have different masses. The setup is as follows. We have a source, which emits atoms. We then ionize the atoms. Then the atoms fly through a capacitor inside a magnetic field. We expect, with $\mathbf{E} = E\mathbf{e}_y$ and $\mathbf{B} = B\mathbf{e}_y$ and $\mathbf{v} = v\mathbf{e}_z$ as starting velocity, that the Lorentz force is

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) = q(E\mathbf{e}_y - vB\mathbf{e}_x).$$

So the equations of motion in the $y$ direction is

$$m\ddot{y} = qE => y = \frac{qE}{2m}t^2.$$

For $x$, assuming the length of is small, we have

$$m\ddot{x} = -qvB \Rightarrow x = -\frac{qvB}{2m}t^2.$$

The time spent in the field is $t = \frac{L}{v}$. Hence,

$$y = \frac{qE}{2m}\frac{L^2}{v^2}, \quad x = -\frac{qvB}{2m}\frac{L^2}{v^2} = -\frac{qBL^2}{2mv}.$$

Hence,

$$\frac{1}{v} = -\frac{x^2 m}{qBL^2} \Rightarrow y = \frac{2Ex^2}{B^2L^2} \cdot \frac{m}{q}.$$

Hence, we expect a parabolic curve. Also, every ion with the same $\frac{m}{q}$ ratio will land on the same parabola. The speed only sets where it lands. If its fast, its near the zero point, if its slow, it lands further away. This was used by Thomson in 1913 to discover the first isotopes of stable elements. One problem of the method is that you have a parabolic curve, which means the intensity per point is low. Ideally, we want to have all $\frac{q}{m}$ ions to land on the same point.

The idea here is to use a radial **E**-field. The ion will the follow a circle.

$$r = \frac{mv^2}{qE}.$$

The magnetic field is then dispersive in momentum with

$$r = \frac{mv}{qB}.$$

So we can do direction focusing and velocity focusing. So now we can determine the mass very well. What is now interesting is the size and charge distribution of atoms.

Lec 9

# 4 Statistical Mechanics

The goal of statistical mechanics is to connect microscopic descriptions to macroscopic variables. For example, we might have a lot of particles in a gas, each with its own position and velocity. We want to connect this to the temperature of the gas.

At the microscopic level, we have deterministic equations of motion. So if we know the position $\mathbf{q}_i$ and momentum $\mathbf{p}_i$ of each particle, we can use Newton's laws to predict the future values of $\mathbf{q}_i$ and $\mathbf{p}_i$ for all time. The issue is, that for a gas at room temperature and pressure, we have $N_p \sim 10^{23}$ particles. This is too much information to store and manipulate. So we have to ask ourselves how to predict bulk properties of the gas from the microscopic equations. To do this, we use statistical methods.

Let us first look at discrete probability distributions. The most famous example is probably the dice. It has 6 outcomes, each with probability 1/6. Another more complex example is a coin toss of 4 fair coins. There are $2^4 = 16$ outcomes. However we only care about the total amount of heads. So we can group the outcomes into 5 groups: 0 heads, 1 head, 2 heads, 3 heads, and 4 heads. The probability of getting 0 heads is 1/16, the probability of getting 1 head is 4/16, the probability of getting 2 heads is 6/16, the probability of getting 3 heads is 4/16, and the probability of getting 4 heads is 1/16. This is called the binomial distribution.

For continuous probability distributions, the probability to find a particle between $x$ and $x+dx$ is given by $p(x)dx$. Note, that $p(x)$ is not a probability! It is a probability density, which becomes a probability when multiplied with an interval.

An important quantity is the EXPECTATION VALUE of a quantity. For example, a dice has the outcomes $X = 1, 2, 3, 4, 5, 6$ with probabilities $p(X) = 1/6$. The expectation value is given by

$$\langle x \rangle = \frac{1 + 2 + \cdots + 6}{6} = \sum XP(X) = 3.5.$$

In the continuous case, the expectation value is given by

$$\langle x \rangle = \frac{\int x \cdot (p(x)\mathrm{d}x)}{\int p(x)\mathrm{d}x}.$$

Consider a single particle in a box. Suppose the particle moves with a velocity $v_z$ to wards a wall. The particle will hit the wall and bounce back. For this the wall excerts a force $F_z(\text{wall})$. The impulse of collision relates to

$$\delta p = -2mv_z.$$

We assume collisions to be repetitive at time intervals $\Delta t = \frac{2L}{v_z}$. The average force on the wall is then given by

$$\langle F_z(\text{wall}) \rangle = \frac{\delta p}{\Delta t} = -\frac{mv_z^2}{L}.$$

In a gas, we have atoms with velocity between $v_z$ and $v_z + \mathrm{d}v_z$, assigned to the probability $p(v_z)\mathrm{d}v_z$. The average force on the wall is then given by

$$\left\langle F^{\text{tot}} \right\rangle = N \int \frac{mv_z^2}{L} p(v_z)\mathrm{d}v_z = \frac{N \cdot m}{L} \langle v_z^2 \rangle.$$

Since $F = pA$, we can write this as

$$pA \cdot L = Nm\langle v_z^2 \rangle \Rightarrow pV = Nm\langle v_z^2 \rangle.$$

We want to use statistical physics to show that

$$\frac{k_B T}{2} = \frac{m\langle v^2 \rangle}{2}.$$

**Definition 4.1: Macrostate**

Macrostates are defined by bulk variables such that $P, V, E, N$. Also probability distributions are macrostates.

**Definition 4.2: Microstate**

Microstates are a description the microscopic variables such that $\mathbf{q}_i, \mathbf{p}_i$ for all particles $i = 1, \ldots, N$.

Returning to our coin example with 4 coins. We have $N_{\text{coins}} = 4$. Each coin is either heads or tails. A microstate is for example: "Coin 1 is heads". A macrostate might be "1 coin is heads". Notice, that the macrostate does not tell us which coin is heads. There are many microstates that correspond to the same macrostate. For example, "Coin 1 is heads" and "Coin 2 is heads" are two different microstates that correspond to the same macrostate of "1 coin is heads".

A different example is a system of particles, which take energies $n \cdot \varepsilon$. Given 7 particles at total energy $3\varepsilon$, we can have for example the two microstates:

$$\text{Microstate 1: } (0, 0, 0, 0, 0, 3\varepsilon, 0)$$
$$\text{Microstate 2: } (0, 0, \varepsilon, 0, \varepsilon, \varepsilon, 0).$$

In this course we will define these as separate macrostates as the probability distributions are different.

## 4.1 The Boltzmann Distribution

We want to look at equilibrium distributions. For this we go back to the coin-flipping game. The coins start in some initial state. At every step of the game, choose one coin at random and flip it. If we interrupted the game after a long time, what can someone predict about the macroscopic variable "Number of heads"?

Consider the macrostate $N_H = 4$. This gives us the probabilistic condition

$$P(3|4) = 1.$$

Since regardless of the microstate, we will always have 4 heads. If we start with $N_H = 3$, we get the probabilistic values

$$P(4|3) = \frac{1}{4}, \quad P(2|3) = \frac{3}{4}.$$

So now we can write down the change in probability of having 4 heads at time $t+1$ given the probabilities at time $t$:

$$\begin{aligned} P_{t+1}(N_H = 4) &= P(4|4)P_t(N_H = 4) + P(4|3)P_t(N_H = 3) \\ \Rightarrow \Delta P(4) &= P_{t+1}(N_H = 4) - P_t(N_H = 4) \\ &= (P(4|4) - 1)P_t(4) + P(4|3)P_t(3) \\ &. \end{aligned}$$

We expect, that after a long time, we reach equilibrium, where the probability distributions become static. Hence $\Delta P_\infty(4) = 0$. This gives us the equation

$$-\left(P(4|4) - 1\right) P_\infty(4) = P(4|3) P_\infty(3)$$
$$P_{\mathrm{SS}}(4) = \frac{1}{4} P_{\mathrm{SS}}(3).$$

We could do the same principle for all other macrostates. Inteterstingly, the probability to find the system in a macrostate with $N_H$ heads is proportional to the amount of microstates that correspond to this macrostate.

This leads us to the FIRST ERGODIC POSTULATE for equilibrium statistical mechanics:

> For a closed system, every microstate which satisfies the global constraints is equally likely to be occupied. - Boltzmann

The second Ergodic postulate states:

> The macrostate which is occupied in thermal equilibrium is the one with the largest number of microstates.

Note that this macrostate must satisfy the fixed constraints (for example the total amount of coins).

One concept following from this is IRREVERSINILITY. If we have a ordered macrostate, i.e. one with only a single microstate, then the system will evolve to a disordered macrostate, i.e. one with many microstates.

### 4.1.1 Closed System

Consider a system of a fixed number of particles $N$ and a fixed total energy $E$. The atoms cannot interact with the environment in any way. We want to count in how many ways we can distribute the energy $E$ among the $N$ particles. For this we make boxes for the energies $\varepsilon_i$ and put $n_i$ particles in each box. We have $\frac{N!}{n_1!(N-n_1)!}$ ways to put $n_1$ particles in the first box. For the second box we then get $\frac{(N-n_1)!}{n_2!(N-n_1-n_2)!}$ ways to put $n_2$ particles in the second box. Continuing this way, we get the total number of microstates

$$\Omega = \frac{N!}{n_1! n_2! \dots n_K!}.$$

# Index

Bragg scattering, 12
Brewster angle, 7

circular polarisation, 6
Compton scattering, 12
Compton wavelength, 12

dynodes, 11

envelope, 9
expectation value, 14

Fabry-Perot etalon, 9
filament, 11
Finesse, 10
first Ergodic postulate, 15
focal length, 2
focal plane, 2
focal point, 2
Fraunhofer approximation, 4
Free spectral range, 10
free-space wavelength, 1
Full width half maximum, 10

grating, 7
Grating Spectrometer, 7

Huygen's principle, 4

irreversinility, 15

lens equation, 2
linear polarisation, 6

magnification, 2
Michelson Spectrometer, 8

optical path, 1

p-polarisation, 6
parallele Polarisation, 6
paraxial approximation, 2
photon, 11
phototube, 11
Planck constant, 11
Planck experiment, 11

quanta, 11

Rayleigh criterion, 5

s-polarisation, 6
senkrechte Polarisation, 6

thermionic emission, 11

work function, 11